Radical expressions

1

a $\sqrt{27}$ b $\sqrt{32}$ c $\sqrt[3]{40}$ d $\sqrt[3]{686}$

2

a $\sqrt{25 \cdot 2} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}$ b $\sqrt{16 \cdot 25} = \sqrt{16} \cdot \sqrt{25} = 4 \cdot 5 = 20$

3

a 27 b $7\sqrt{10}$ c $5\sqrt{6}$ d 8

e 1 f -1 g -4 h 2

i $2\sqrt[3]{6}$ j $3\sqrt[3]{3}$ k 35 l $\frac{1}{2}$

m 3 n $3\sqrt[3]{4}$

4

a Yes b No c Yes

5

a x^4 b m c v^6 d $6x^3$

e bx^3 f n^3p^3 g $9x^6y^4$ h $7y^2$

i a j b^7 k $4x^4$ l x^4y^5

m jx^2 n $(kx)^4$ o $5x^7y^{10}$ p $2ay^2\sqrt{7a}$

q m^2 r $8\sqrt{x}$

6

a $\sqrt[3]{4}$ b $\frac{x^2}{2}$ c $\frac{7x^2}{8}$ d $\frac{6x^9}{y^{10}}$

7

a $2x+9$ b $4y+3$ c $y\left(x+9\right)$

8

a 5 b 6 c 5

9

a -15 b -2 c 20

10

a False

b True

c False

11 $x + 3$

12 $k = 6$